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一.

$$\because f(x,y)=\begin{cases} C\cdot e^{-(2x+y)} & x>0,\ y>0 \\ 0 & \text{else} \end{cases}$$

$$\therefore \int_0^{\infty}\int_0^{\infty} C\cdot e^{-(2x+y)}\,dx\,dy = \frac{C}{2}=1; \qquad \therefore C=2.$$

$$\therefore f_{X|Y}(x|y)=\frac{f(x,y)}{f(y)}=\frac{C\cdot e^{-(2x+y)}}{\int_0^{\infty} Ce^{-(2x+y)}\,dx}$$

$$\therefore f_{X|Y}(x|y)=\begin{cases} 2e^{-2x} & x>0 \\ 0 & x\le 0 \end{cases}$$

二.

(a) $$f(x|y)=\begin{cases} \frac{f(x,y)}{f(y)}=\lambda e^{-\lambda x} & x>0 \\ 0 & \text{其它} \end{cases}$$

(b) $$P(Z=1)=\int_0^{\infty} dy\int_0^{Y} f(x,y)\,dx = 1-\frac{1}{\frac{\lambda}{\mu}+1}=\frac{\lambda}{\lambda+\mu}$$

$$P(Z=0)=\frac{\mu}{\lambda+\mu}$$

Z	0	1
P	$\frac{\mu}{\lambda+\mu}$	$\frac{\lambda}{\lambda+\mu}$

$$\therefore E(Z)=\frac{\lambda}{\lambda+\mu}$$

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三.

$\because f(x,y)=\begin{cases}\lambda^2 e^{-\lambda x} & 0<y<x \\ 0 & else\end{cases}$

$\therefore E(X|Y=y)=\int_y^{+\infty} x p(x|y)\,dx$

$=\begin{cases}y+\frac{1}{\lambda} & (y>0) \\ 0 & y\le 0\end{cases}$

$D(X|Y=y)=E(x^2|Y=y)-E^2(X|Y=y)=\begin{cases}\frac{1}{\lambda^2} & y>0 \\ 0 & y\le 0\end{cases}$

四. $p(x,\eta)=\frac{1}{a(a-\eta)}$

$p(\eta)=\frac{1}{a}$

$p(x|\eta)=\int_\eta^a \frac{x}{a-\eta}dx=\frac{a+\eta}{2}$

$\because \eta\sim U[0,a]$

$\therefore E(X|\eta)=\frac{E(X,\eta)}{E(\eta)}\sim U[\frac{a}{2},a]$

五. $E(Y)E(X|Y)=E(X,Y)=E(X)\cdot E(Y)$

$P(Y)P(X|Y)=P(XY)$

$\therefore E(X^2|X)=X^2\neq E(X^2)$

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六. $\because X_1、X_2$ 独立

$\therefore f(x_1)=\lambda_1 e^{-\lambda_1 x_1}$

$f(x_2)=\lambda_2 e^{-\lambda_2 x_2}$

$\therefore P(X_1 \mid X_1<X_2)=\int_{x_1}^{+\infty} f(x_1,x_2)\,dx_2 = \lambda_1 e^{-(\lambda_1+\lambda_2)x_1}$

$\therefore E(X_1 \mid X_1<X_2)=\int_0^{+\infty} x_1 P(X_1 \mid X_1<X_2)\,dx_1 = \frac{1}{\lambda_1+\lambda_2}$

七. $\because f(x,y)=\begin{cases} 2e^{-(x+2y)} & x>0,\ y>0 \\ 0 \end{cases}$

又 $\because z=x+2y$

$f(z)=\int_0^z dx \int_0^{\frac{z-x}{2}} 2e^{-(x+2y)}\,dy = 1-e^{-z}-ze^{-z} \quad z>0.$

$\therefore f(z)=\begin{cases} ze^{-z} & z>0 \\ 0 & z\le 0. \end{cases}$

八.

(1) $\because f(x)=\lambda e^{-\lambda x}$

$f(y)=\lambda e^{-\lambda y}$

$\therefore f(x,y)=\lambda^2 e^{-\lambda(x+y)}$

$\therefore F(x+Y)=\begin{cases} \int_0^z dY \int_0^{z-y} \lambda^2 e^{-\lambda} dx = \int_0^z \lambda^2 e^{-\lambda z}\,dz = \lambda^2 z e^{-\lambda z} \\ 0 \end{cases}$

$\therefore f(x+y)=\begin{cases} \lambda^2 z e^{-\lambda z} & z>0 \\ 0 & z<0. \end{cases}$

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九.

$f(x)=e^{-x}$, $x>0$

$f(y)=e^{-y}$, $y>0$.

$\therefore f(u,v)=f(x,y)\cdot\frac{u}{(1+v)^2}=e^{-u}\frac{u}{(1+v)^2}$

又 $\because \operatorname{Cov}(u,v)=\operatorname{cov}(x+y,\frac{x}{y})=E[\frac{x^2}{y}+x-\frac{2x}{y}-x-y+2]=0$, $\therefore u,v$ 独立.

十.

$\because f(x,y)=\begin{cases}\frac{12}{7}x(x+y) & 0<x,y<1\\ 0 & \text{else}\end{cases}$

$\therefore f(u,v)=\frac{12}{7}(2x^2+2xy)=\frac{12}{7}(u^2+v^3+2uv)$

十一.

(a) $(X,Y)\sim N(\mu_1,\mu_2,\rho,\sigma_1^2,\sigma_2^2)$

$\therefore \hat{X}=E(X|Y)=\int_{-\infty}^{+\infty}x\frac{f(x,y)}{f(y)}dx=\mu_1+\frac{\rho\sigma_1}{\sigma_2}(Y-\mu_2)$

(b) $\because \mu_1=\mu_2=0$;

$\therefore E(X^2|Y^2=y^2)=\int_0^{+\infty}x^2 f(x^2|Y^2=y^2)\,dx=\int_0^{+\infty}x^2\frac{f(x^2,Y^2=y^2)}{f(Y^2=y^2)}dx$

$=\frac{1}{2}\int_0^{+\infty}x^2\frac{f(x^2,Y=y)+f(x^2,Y=-y)}{f(Y=y)}$

$=\frac{1}{2}E(x^2|Y=y)+\frac{1}{2}E(x^2|Y=-y)$

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(C) $E(X^2)=1$, $E(Y^2)=1$, $\mathrm{cov}(X^2,Y^2)=\rho^2$

则由(a)可知, $E(X^2\mid Y^2=y^2)=1+\rho^2(y^2-1)$

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